

CÁNCER DE PULMÓN OHIO

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Intriducción

Este conjunto de datos se deriva de un estudio de mortalidad por cáncer de pulmón en Ohio realizado entre 1969 y 1971 a nivel de condado. La base de datos **ohiolung** contiene información para los 88 condados del estado de Ohio, (EE. UU.). La base de datos proviene del GeoDa (Center for Geospatial Analysis and Computation) y presenta 42 variables. En las estas se presentan datos referentes a la mortalidad por cáncer de pulmón estratificada por género (masculino/femenino) y raza (blanca/negra), junto variables geográficas básicas.

La siguiente tabla resume las variables principales de la base de datos:

Table 1: Definición de variables principales en la base de datos
ohlung.shp

Variable	Descripcion
CountyID	ID secuencial del condado (orden alfabético)
NAME	Nombre del condado
FIPSNO	Código FIPS del condado (numérico)
AREA	Área del polígono (condado)
PERIMETER	Perímetro del polígono (condado)
RECORD_ID	ID único del registro
COUNTYID	ID del condado
LGRyy	Casos de cáncer de pulmón para género G (M/F) y raza R (W/B) en año yy (1968, 1978, 1988)
POPGRyy	Población en riesgo para género G y raza R en año yy
LGyy	Total de casos de cáncer de pulmón por género G en año yy
POPGyy	Población total en riesgo por género G en año yy

Table 2: Vista de los datos en formato largo

COUNTYID	NAME	AREA	PERIMETER	gender	race	year	cases	pop
48	Lucas	873382000	164533	Male	White	1968	128	205421
48	Lucas	873382000	164533	Female	White	1968	27	220106
48	Lucas	873382000	164533	Male	Black	1968	19	25956
48	Lucas	873382000	164533	Female	Black	1968	1	28527
48	Lucas	873382000	164533	Male	White	1978	149	194976
48	Lucas	873382000	164533	Female	White	1978	48	210307
48	Lucas	873382000	164533	Male	Black	1978	34	30252

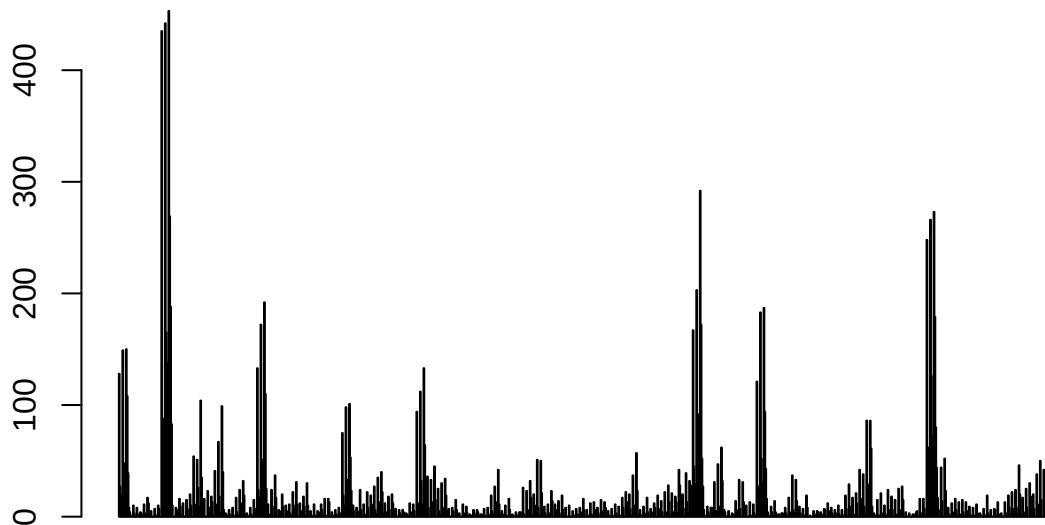
COUNTYID	NAME	AREA	PERIMETER	gender	race	year	cases	pop
48	Lucas	873382000	164533	Female	Black	1978	4	33790
48	Lucas	873382000	164533	Male	White	1988	150	185367
48	Lucas	873382000	164533	Female	White	1988	108	200458

```
ohlung_long %>%
  dplyr::select(cases) %>%
  dplyr::filter(cases == 0) %>%
  dplyr::summarise(count = n())
```

```
## # A tibble: 1 x 1
##   count
##   <int>
## 1    386
```

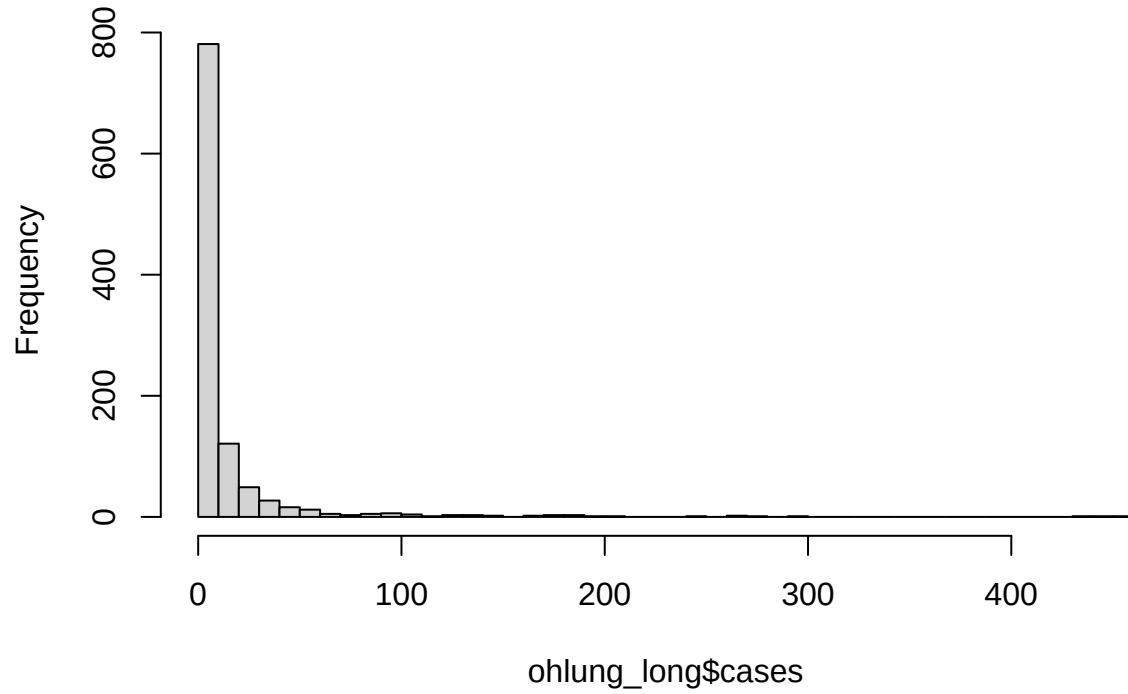
Notese que tenemos de 1056 observaciones, 386 son 0 casos, esto puede representar un problema más adelante en los modelos, ya que puede tender a inflar la varianza.

```
barplot(ohlung_long$cases)
```

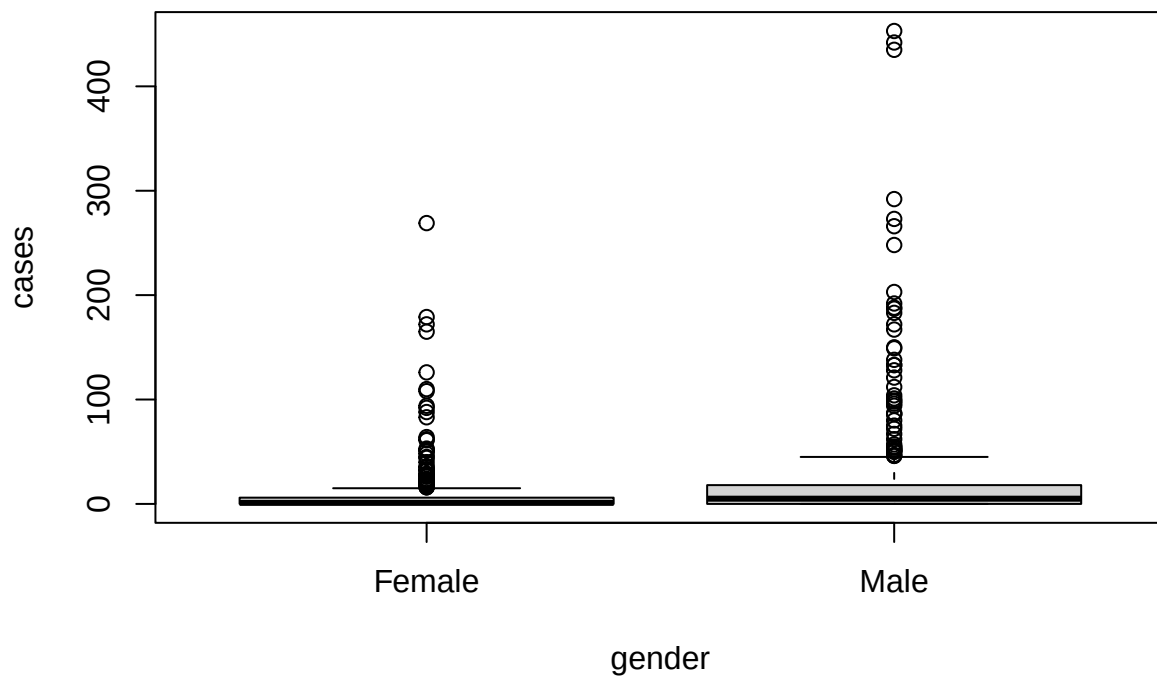


```
hist(ohlung_long$cases, breaks = 50)
```

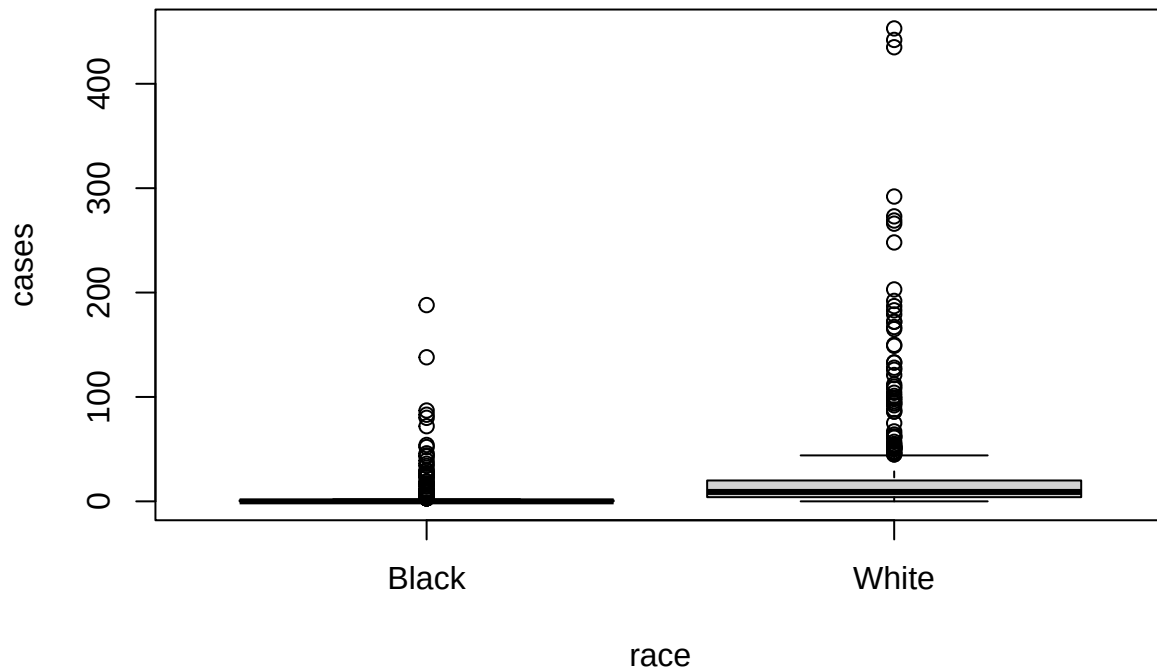
Histogram of ohlung_long\$cases



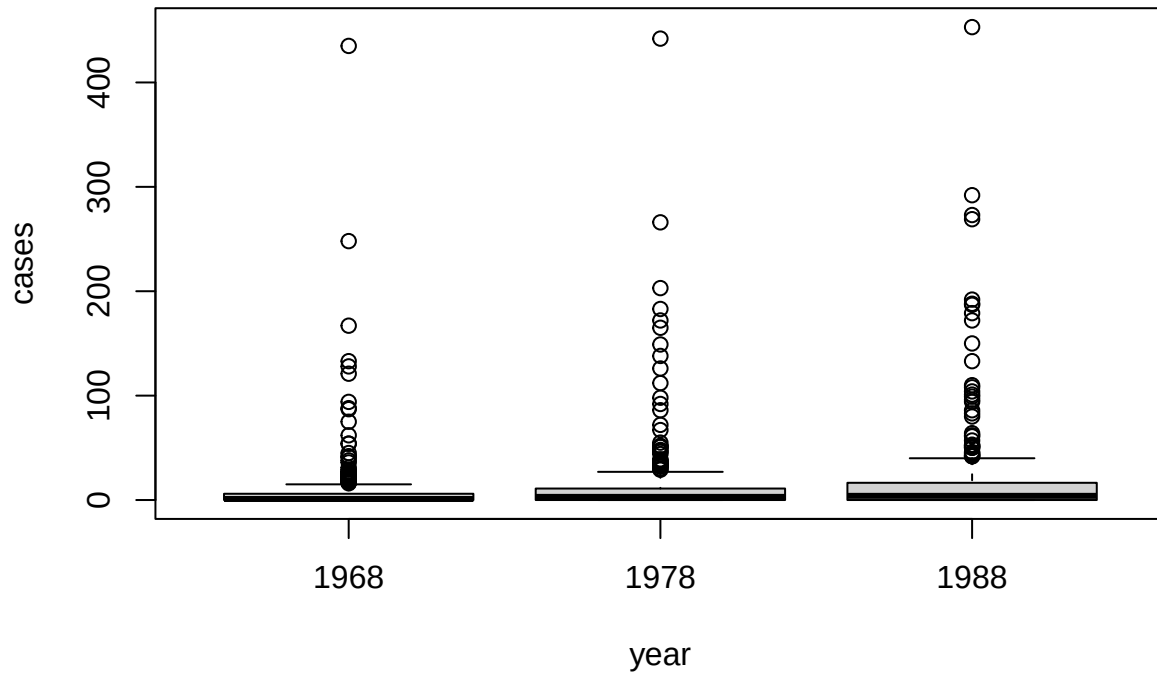
```
boxplot(cases ~ gender, data=ohlung_long)
```



```
boxplot(cases ~ race, data=ohlung_long)
```

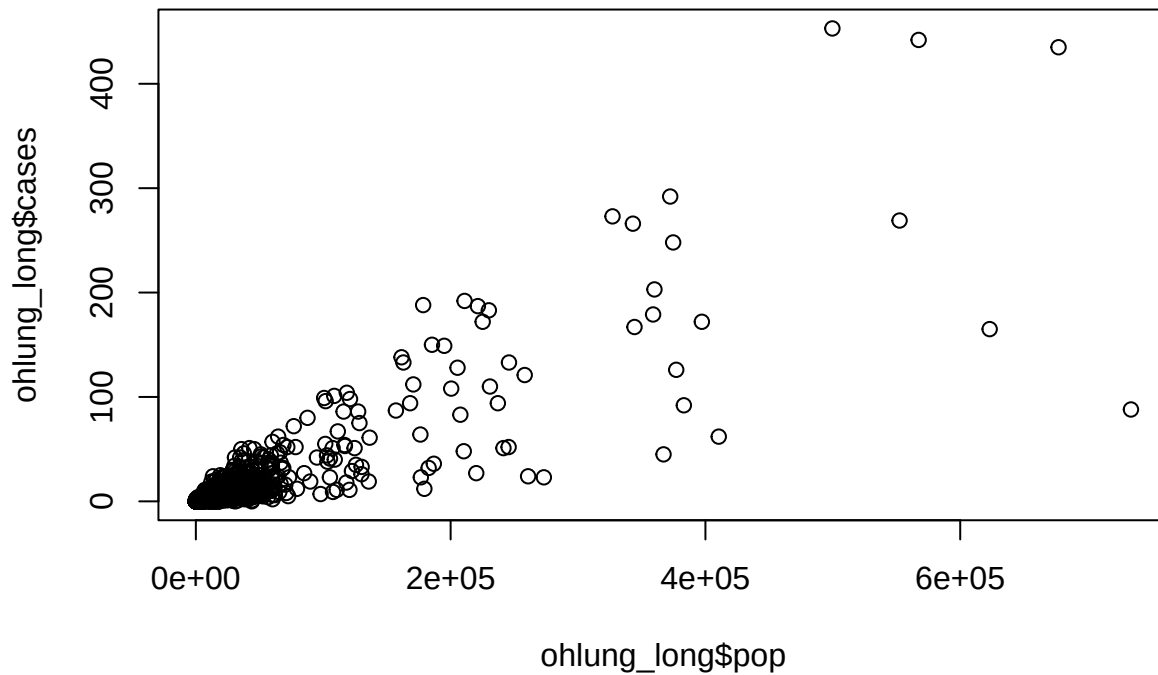


```
boxplot(cases ~ year, data=ohlung_long)
```



Presencia de datos atípicos por años, mas dispersión conforme pasan los años, sin embargo la media parece mantenerse. Más dispersos en hombres que mujeres pero media constante, más datos atípicos por hombres que por mujeres.

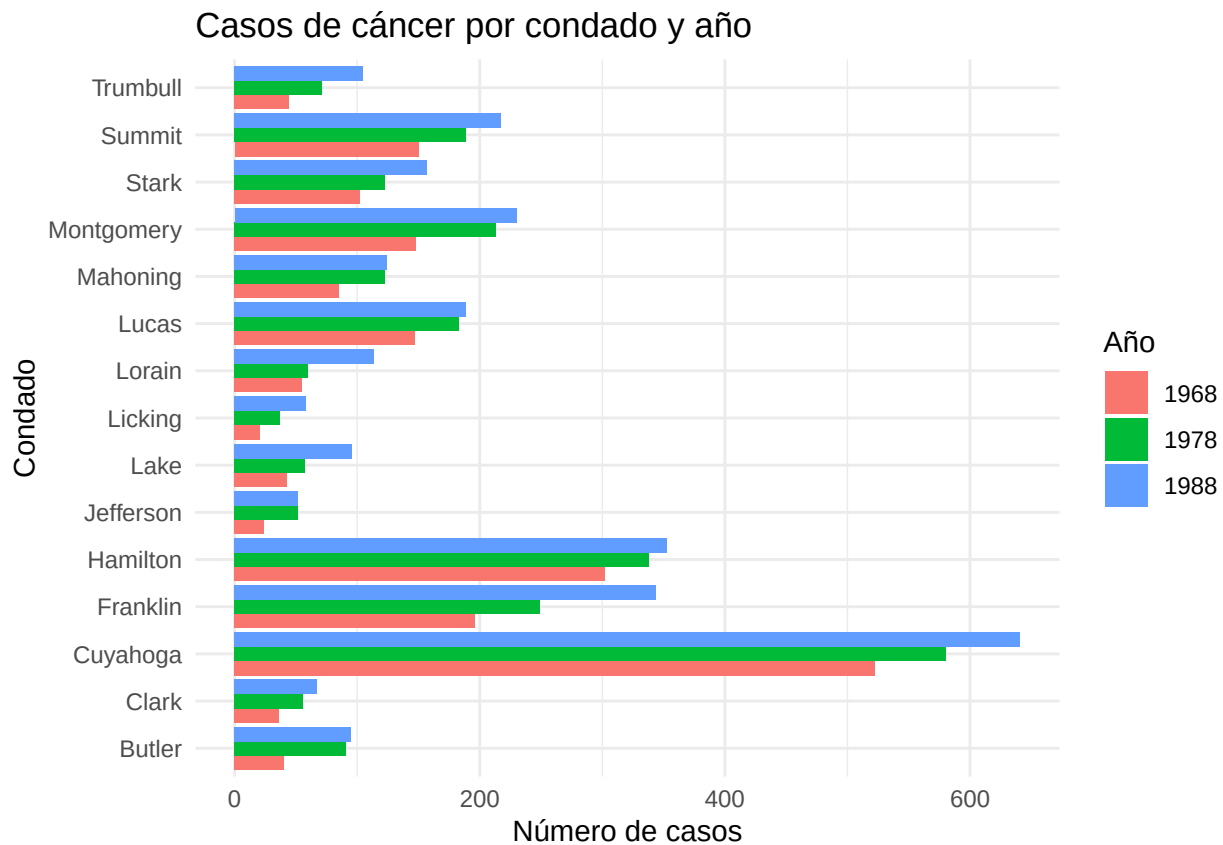
```
plot(ohlung_long$pop, ohlung_long$cases)
```



```
casos_condado <- ohlung_long %>%
  group_by(NAME, year, gender) %>%
  summarise(casos = sum(cases, na.rm = TRUE), .groups = "drop")

top_condados <- ohlung_long %>%
  group_by(NAME) %>%
  summarise(total_casos = sum(cases, na.rm = TRUE)) %>%
  slice_max(total_casos, n = 15)

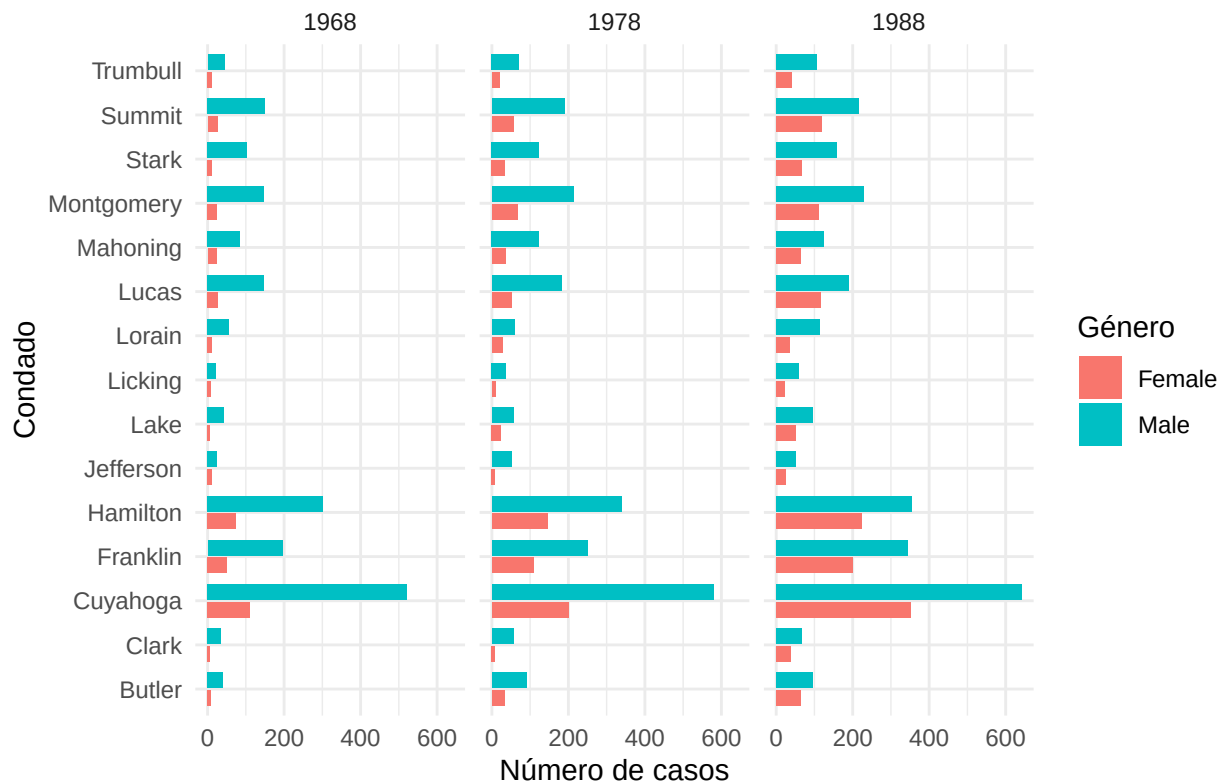
ggplot(
  casos_condado %>% filter(NAME %in% top_condados$NAME),
  aes(
    x = NAME,
    y = casos,
    fill = factor(year)
  )
) +
  geom_col(position = "dodge") +
  coord_flip() +
  labs(
    title = "Casos de cáncer por condado y año",
    x = "Condado",
    y = "Número de casos",
    fill = "Año"
  ) +
  theme_minimal()
```



Podemos ver que por condado va aumentando los casos cada año evaluado.

```
ggplot(
  casos_condado %>% filter(NAME %in% top_condados$NAME),
  aes(
    x = NAME,
    y = casos,
    fill = gender,
    group = interaction(year, gender)
  )
) +
  geom_col(
    position = position_dodge2(preserve = "single"),
    width = 0.8
  ) +
  coord_flip() +
  facet_grid(~ year) +
  labs(
    title = "Casos de cáncer por condado, año y género",
    x = "Condado",
    y = "Número de casos",
    fill = "Género"
  ) +
  theme_minimal()
```

Casos de cáncer por condado, año y género



```
top_condados2 <- ohlung_long %>%
  group_by(NAME) %>%
  summarise(total_casos = sum(cases, na.rm = TRUE),
            area = first(AREA)) %>%
  slice_max(total_casos, n = 15)
knitr::kable(top_condados2, caption = "Top 15 condados con más casos")
```

Table 3: Top 15 condados con más casos

NAME	total_casos	area
Cuyahoga	2407	1242480000
Hamilton	1435	1084790000
Franklin	1146	1442010000
Montgomery	791	1210680000
Summit	759	1093150000
Lucas	715	873382000
Stark	495	1499940000
Mahoning	453	1077280000
Butler	330	1214420000
Lorain	303	1306670000
Trumbull	293	1635690000
Lake	278	611960000
Clark	211	1030010000
Jefferson	172	1032610000

NAME	total_casos	area
Licking	157	1779800000

```
numericas <- ohlung_long %>% dplyr::select(-c(COUNTYID,NAME,gender,race,year))
cor(numericas)
```

```
##           AREA PERIMETER      cases      pop
## AREA      1.00000000 0.64757574 0.01423795 0.02560406
## PERIMETER 0.64757574 1.00000000 0.08427196 0.08279894
## cases     0.01423795 0.08427196 1.00000000 0.84792297
## pop       0.02560406 0.08279894 0.84792297 1.00000000
```

```
pcor(numericas)
```

```
## $estimate
##           AREA PERIMETER      cases      pop
## AREA      1.00000000 0.648561827 -0.04116936 0.015242745
## PERIMETER 0.64856183 1.000000000 0.04694801 0.006456732
## cases     -0.04116936 0.046948014 1.00000000 0.846666565
## pop       0.01524275 0.006456732 0.84666657 1.000000000
##
## $p.value
##           AREA PERIMETER      cases      pop
## AREA      0.000000e+00 7.802003e-127 1.816936e-01 6.210923e-01
## PERIMETER 7.802003e-127 0.000000e+00 1.277047e-01 8.341566e-01
## cases     1.816936e-01 1.277047e-01 0.000000e+00 1.691562e-290
## pop       6.210923e-01 8.341566e-01 1.691562e-290 0.000000e+00
##
## $statistic
##           AREA PERIMETER      cases      pop
## AREA      0.000000 27.6364444 -1.336443 0.4944490
## PERIMETER 27.636444 0.0000000 1.524419 0.2094256
## cases     -1.336443 1.5244186 0.000000 51.6068802
## pop       0.494449 0.2094256 51.606880 0.0000000
##
## $n
## [1] 1056
##
## $gp
## [1] 2
##
## $method
## [1] "pearson"
```

Alta correlación entre casos y población, se presenta correlación moderadamente alta entre el área y el perímetro. En las correlaciones parciales podemos ver que la relación entre las variables independientes, área y perímetro, y la variable dependiente casos, tiene una correlación baja, sin embargo llama la atención el hecho de que con área maneja una correlación lineal inversa, mientras que con perímetro maneja una correlación lineal directa.


```
glm1 <- glm(cases~PERIMETER+year+gender+race+NAME+pop,
            data=ohlung_long,family = poisson(link = "log"))
summary(glm1)
```

```
##
## Call:
## glm(formula = cases ~ PERIMETER + year + gender + race + NAME +
##      pop, family = poisson(link = "log"), data = ohlung_long)
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.503e+00  9.434e-01  -1.594 0.111044
## PERIMETER    -2.133e-06  6.127e-06  -0.348 0.727753
## year1978      3.474e-01  2.280e-02  15.238 < 2e-16 ***
## year1988      6.183e-01  2.180e-02  28.359 < 2e-16 ***
## genderMale    9.205e-01  1.875e-02  49.089 < 2e-16 ***
## raceWhite     2.656e+00  4.488e-02  59.187 < 2e-16 ***
## NAMEAllen     1.412e+00  1.515e-01   9.323 < 2e-16 ***
## NAMEAshland   5.355e-01  1.812e-01   2.956 0.003117 **
## NAMEAshtabula 1.535e+00  2.070e-01   7.416 1.20e-13 ***
## NAMEAthens    7.177e-01  2.265e-01   3.168 0.001532 **
## NAMEAuglaize  1.788e-01  2.184e-01   0.819 0.412944
## NAMEBelmont   1.377e+00  1.478e-01   9.313 < 2e-16 ***
## NAMEBrown     4.045e-01  2.739e-01   1.477 0.139676
## NAMEButler    2.534e+00  1.361e-01  18.615 < 2e-16 ***
## NAMECarroll   -4.940e-01  2.495e-01  -1.980 0.047707 *
## NAMEChampaign 1.564e-01  2.074e-01   0.754 0.450733
## NAMEClark     1.915e+00  1.640e-01  11.680 < 2e-16 ***
## NAMEClermont  1.626e+00  1.434e-01  11.339 < 2e-16 ***
## NAMEClinton   2.144e-01  2.187e-01   0.980 0.326927
## NAMEColumbiana 1.614e+00  1.701e-01   9.490 < 2e-16 ***
## NAMECoshocton 1.848e-01  2.083e-01   0.887 0.375168
## NAMECrawford  7.332e-01  2.234e-01   3.282 0.001031 **
## NAMECuyahoga  5.874e+00  1.960e-01  29.965 < 2e-16 ***
## NAMEDarke     7.390e-01  1.760e-01   4.198 2.70e-05 ***
## NAMEDefiance  3.849e-02  2.047e-01   0.188 0.850850
## NAMEDelaware  4.454e-01  1.817e-01   2.451 0.014230 *
## NAMEErie      1.195e+00  2.086e-01   5.729 1.01e-08 ***
## NAMEFairfield 1.210e+00  1.766e-01   6.850 7.37e-12 ***
## NAMEFayette   1.037e-02  2.264e-01   0.046 0.963471
## NAMEFranklin  4.517e+00  1.418e-01  31.853 < 2e-16 ***
## NAMEFulton    -2.208e-01  2.456e-01  -0.899 0.368652
## NAMEGallia    3.770e-01  2.316e-01   1.628 0.103494
## NAMEGeauga     4.385e-01  1.867e-01   2.349 0.018841 *
## NAMEGreene    1.321e+00  1.850e-01   7.139 9.37e-13 ***
## NAMEGuernsey  5.462e-01  2.028e-01   2.693 0.007072 **
## NAMEHamilton  4.709e+00  1.441e-01  32.686 < 2e-16 ***
## NAMEHancock   6.349e-01  1.734e-01   3.661 0.000251 ***
## NAMEHardin    1.385e-01  2.027e-01   0.683 0.494480
## NAMEHarrison  -2.811e-01  2.426e-01  -1.159 0.246510
## NAMEHenry     -4.801e-01  2.452e-01  -1.958 0.050259 .
## NAMEHighland  4.098e-01  1.874e-01   2.186 0.028789 *
## NAMEHocking   -7.536e-02  2.279e-01  -0.331 0.740884
```

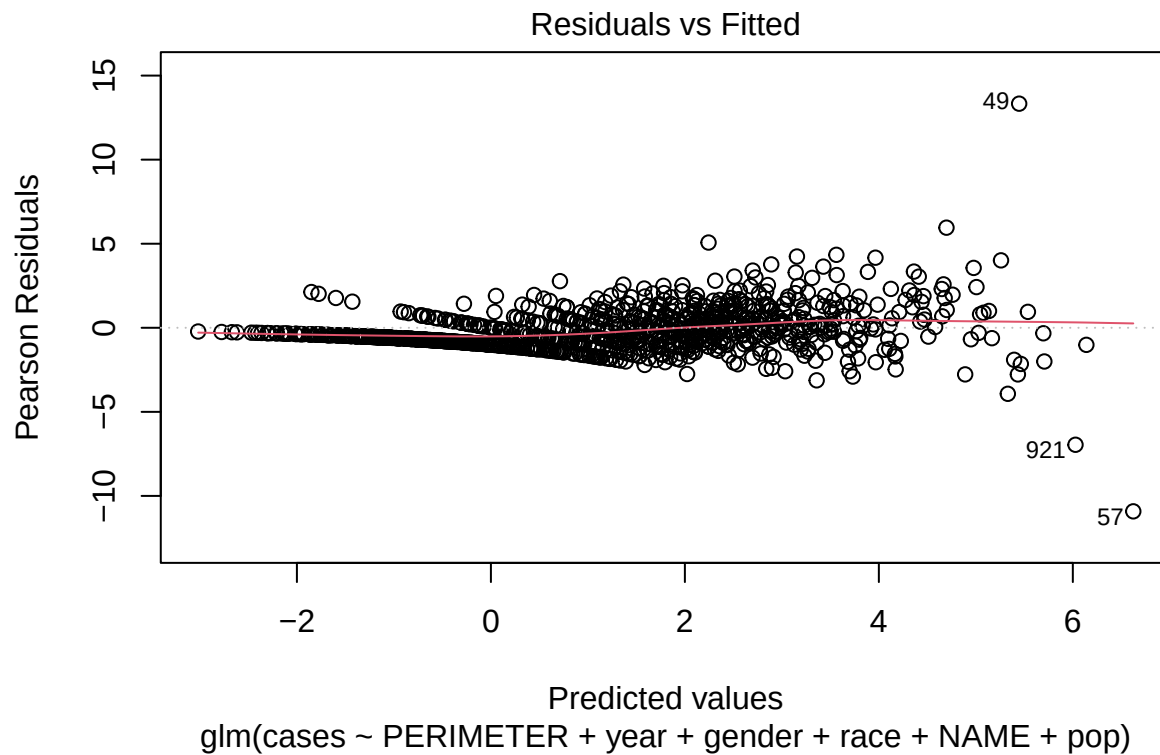
```

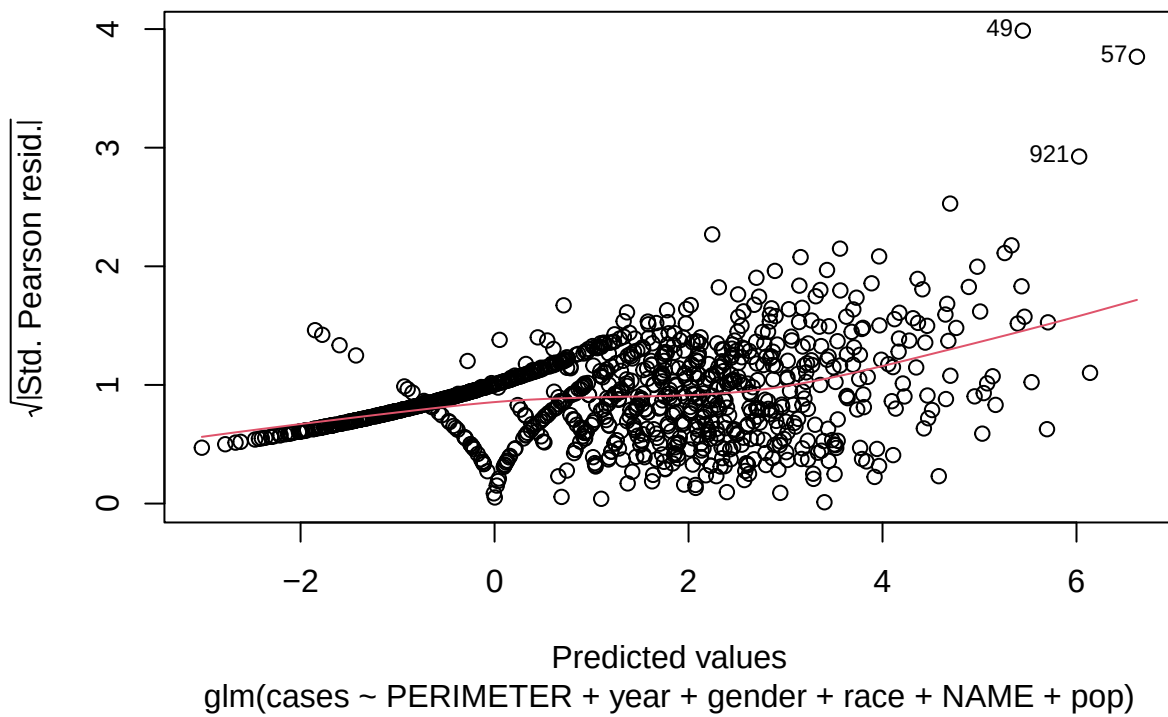
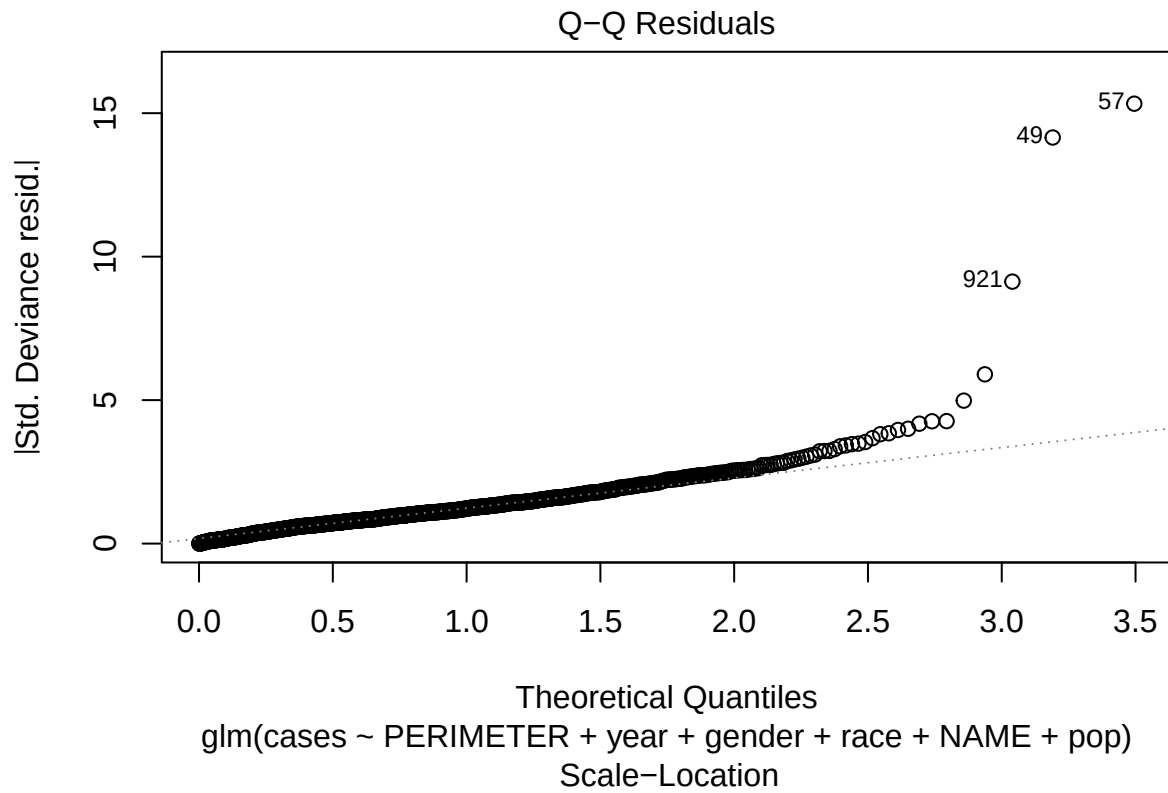
## NAMEHolmes      -8.043e-01  2.798e-01  -2.874  0.004051  **
## NAMEHuron       4.897e-01  1.806e-01   2.711  0.006710  **
## NAMEJackson     3.654e-01  1.962e-01   1.862  0.062608  .
## NAMEJefferson   1.653e+00  1.446e-01  11.433  < 2e-16  ***
## NAMEKnox        6.384e-01  1.968e-01   3.244  0.001178  **
## NAMELake        2.262e+00  2.119e-01  10.674  < 2e-16  ***
## NAMELawrence    1.170e+00  1.992e-01   5.875  4.24e-09  ***
## NAMELicking     1.674e+00  2.084e-01   8.035  9.36e-16  ***
## NAMELogan       5.083e-01  1.860e-01   2.732  0.006286  **
## NAMELorain      2.503e+00  1.703e-01  14.704  < 2e-16  ***
## NAMELucas       3.572e+00  1.474e-01  24.231  < 2e-16  ***
## NAMEMadison     7.510e-02  2.117e-01   0.355  0.722725
## NAMEMahoning    2.862e+00  1.291e-01  22.167  < 2e-16  ***
## NAMEMarion      1.291e+00  1.493e-01   8.648  < 2e-16  ***
## NAMEMedina      1.022e+00  1.651e-01   6.190  6.00e-10  ***
## NAMEMeigs       7.108e-02  2.646e-01   0.269  0.788227
## NAMEMercer      -1.985e-01  2.262e-01  -0.878  0.380193
## NAMEMiami       9.582e-01  1.981e-01   4.836  1.32e-06  ***
## NAMEMonroe      -5.423e-01  2.492e-01  -2.177  0.029514  *
## NAMEMontgomery  3.749e+00  1.355e-01  27.661  < 2e-16  ***
## NAMEMorgan      -6.265e-01  2.612e-01  -2.398  0.016465  *
## NAMEMorrow      -4.776e-01  2.441e-01  -1.956  0.050423  .
## NAMEMuskingum   1.594e+00  2.128e-01   7.491  6.84e-14  ***
## NAMENoble       -8.896e-01  3.372e-01  -2.638  0.008329  **
## NAMEOttawa      3.136e-01  2.369e-01   1.323  0.185677
## NAMEPaulding    -3.142e-01  2.490e-01  -1.262  0.206998
## NAMEPerry       8.669e-02  2.023e-01   0.428  0.668357
## NAMEPickaway    3.568e-01  2.008e-01   1.777  0.075553  .
## NAMEPike        -3.143e-02  2.119e-01  -0.148  0.882060
## NAMEPortage     1.189e+00  1.660e-01   7.166  7.72e-13  ***
## NAMEPreble      -3.345e-03  2.318e-01  -0.014  0.988486
## NAMEPutnam      -3.732e-01  2.364e-01  -1.579  0.114406
## NAMERichland    1.639e+00  1.596e-01  10.269  < 2e-16  ***
## NAMERoss        1.073e+00  2.183e-01   4.915  8.89e-07  ***
## NAMESandusky    8.442e-01  1.664e-01   5.072  3.93e-07  ***
## NAMEScioto      1.525e+00  4.276e-01   3.566  0.000362  ***
## NAMESeneca      9.463e-01  1.619e-01   5.846  5.04e-09  ***
## NAMEShelby      2.279e-01  2.175e-01   1.048  0.294579
## NAMEStark       3.137e+00  1.756e-01  17.868  < 2e-16  ***
## NAMESummit      3.686e+00  1.298e-01  28.403  < 2e-16  ***
## NAMETrumbull    2.409e+00  1.432e-01  16.820  < 2e-16  ***
## NAMETuscarawas  1.244e+00  1.898e-01   6.554  5.59e-11  ***
## NAMEUnion       -3.938e-01  2.377e-01  -1.657  0.097491  .
## NAMEVan Wert    -3.198e-01  2.349e-01  -1.362  0.173312
## NAMEVinton      -1.166e+00  3.311e-01  -3.521  0.000430  ***
## NAMEWarren      1.217e+00  1.763e-01   6.905  5.01e-12  ***
## NAMEWashington  9.633e-01  5.278e-01   1.825  0.067999  .
## NAMEWayne       8.103e-01  1.674e-01   4.842  1.29e-06  ***
## NAMEWilliams    -1.842e-01  2.384e-01  -0.772  0.439895
## NAMEWood        7.268e-01  1.874e-01   3.878  0.000105  ***
## NAMEWyandot     NA          NA          NA          NA
## pop             -3.146e-06  1.683e-07 -18.695  < 2e-16  ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

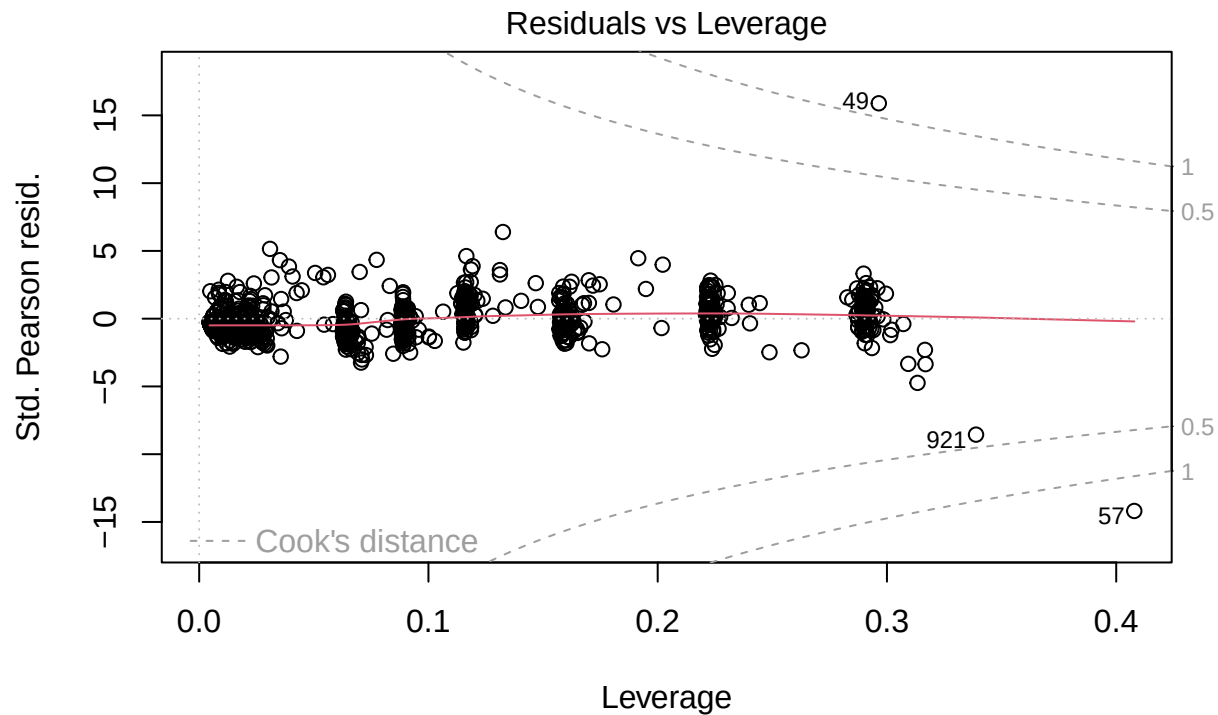
```

```
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 42967.0 on 1055 degrees of freedom
## Residual deviance: 1883.6 on 963 degrees of freedom
## AIC: 4706.2
##
## Number of Fisher Scoring iterations: 5
```

```
plot(glm1)
```







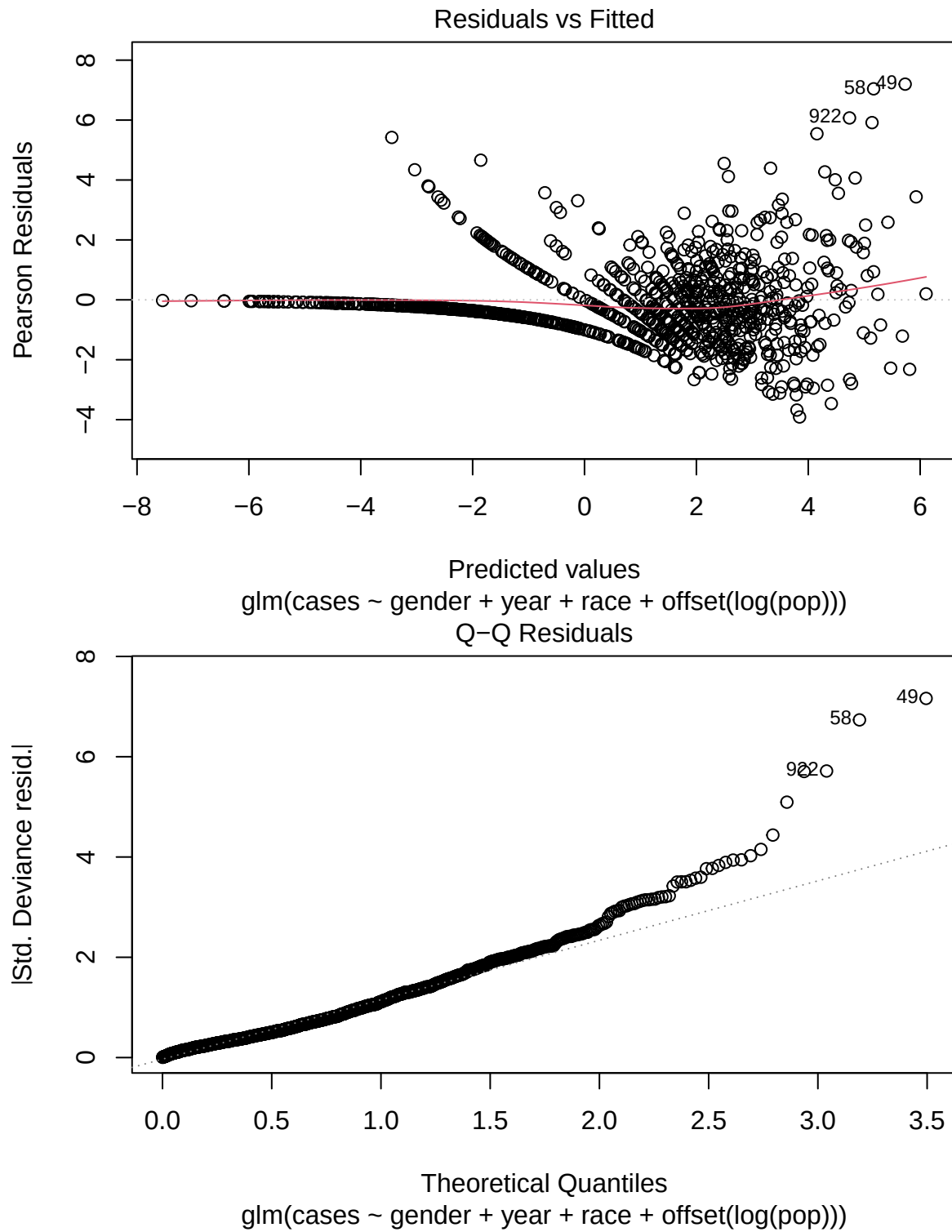
`glm(cases ~ PERIMETER + year + gender + race + NAME + pop)`

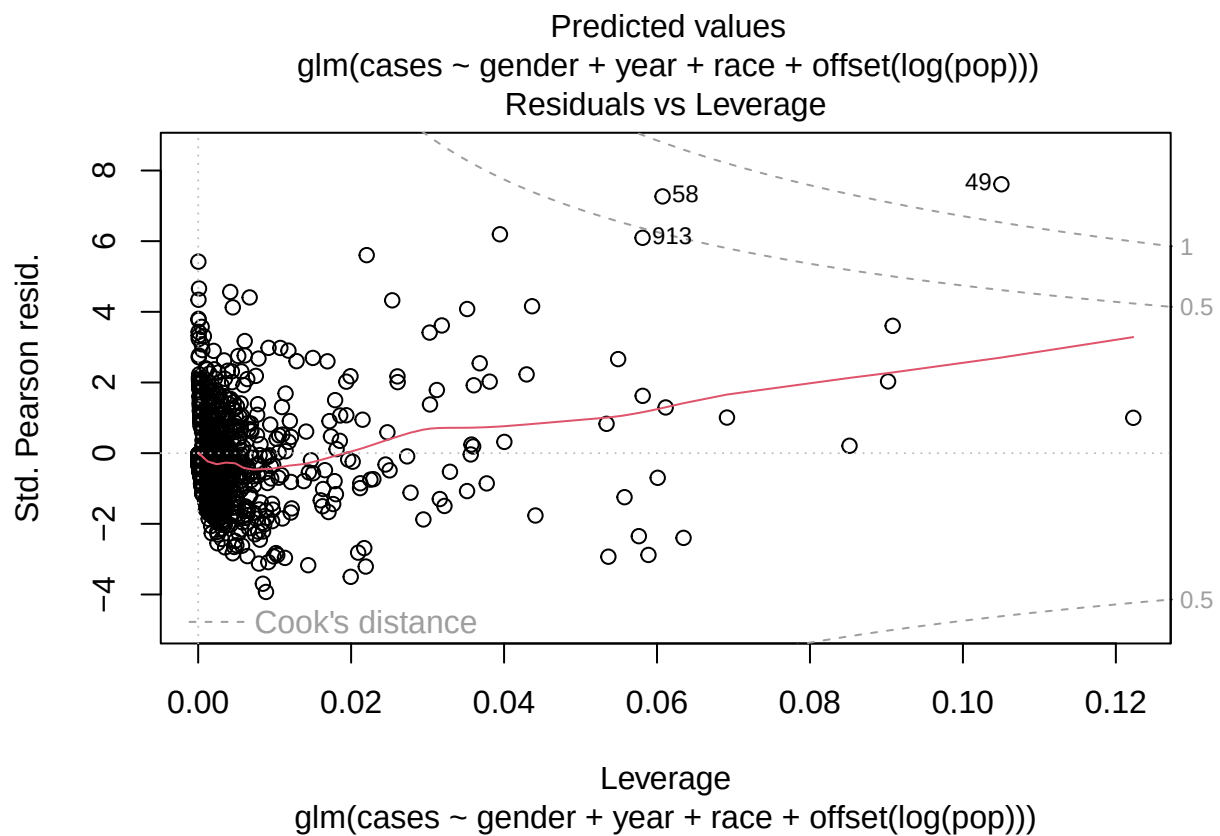
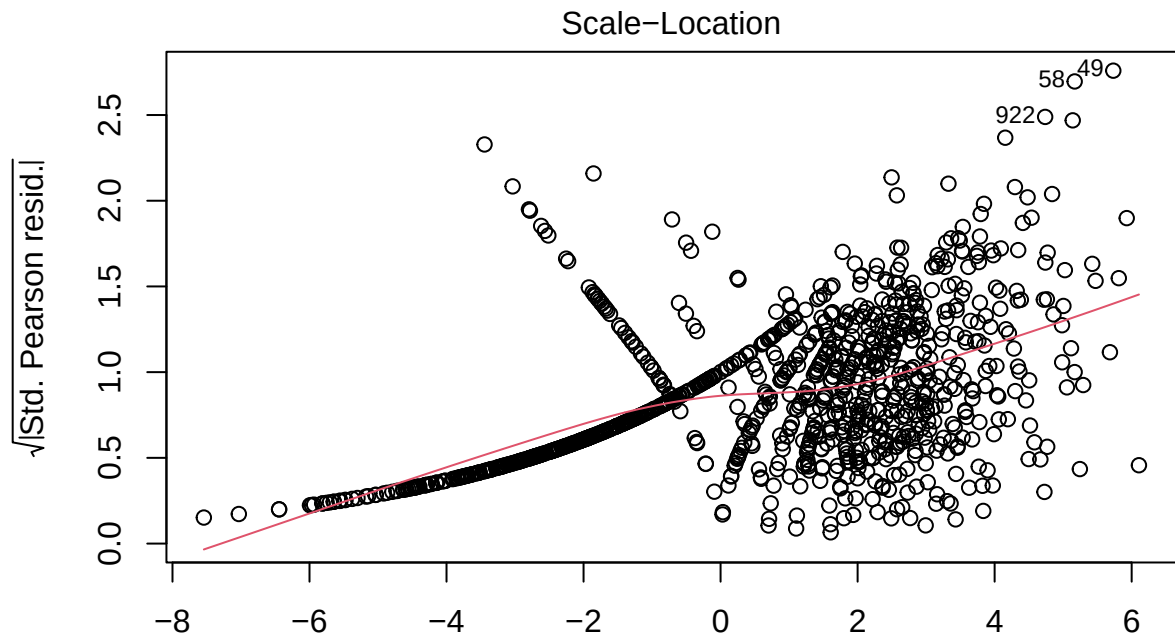
Perimeter no es significativa

```
glm2 <- glm(cases~gender + year + race +
             offset(log(pop)),
             data=ohlung_long,family = poisson(link = "log"))
summary(glm2)
```

```
##
## Call:
## glm(formula = cases ~ gender + year + race + offset(log(pop)),
##      family = poisson(link = "log"), data = ohlung_long)
##
## Coefficients:
##              Estimate Std. Error  z value Pr(>|z|)
## (Intercept) -8.64087    0.03212 -269.026 < 2e-16 ***
## genderMale   1.03832    0.01856  55.942 < 2e-16 ***
## year1978     0.37313    0.02272  16.423 < 2e-16 ***
## year1988     0.67867    0.02149  31.583 < 2e-16 ***
## raceWhite   -0.09129    0.02598  -3.514 0.000442 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 6330.2  on 1055  degrees of freedom
## Residual deviance: 1731.9  on 1051  degrees of freedom
## AIC: 4378.5
##
## Number of Fisher Scoring iterations: 4
```

```
plot(glm2)
```



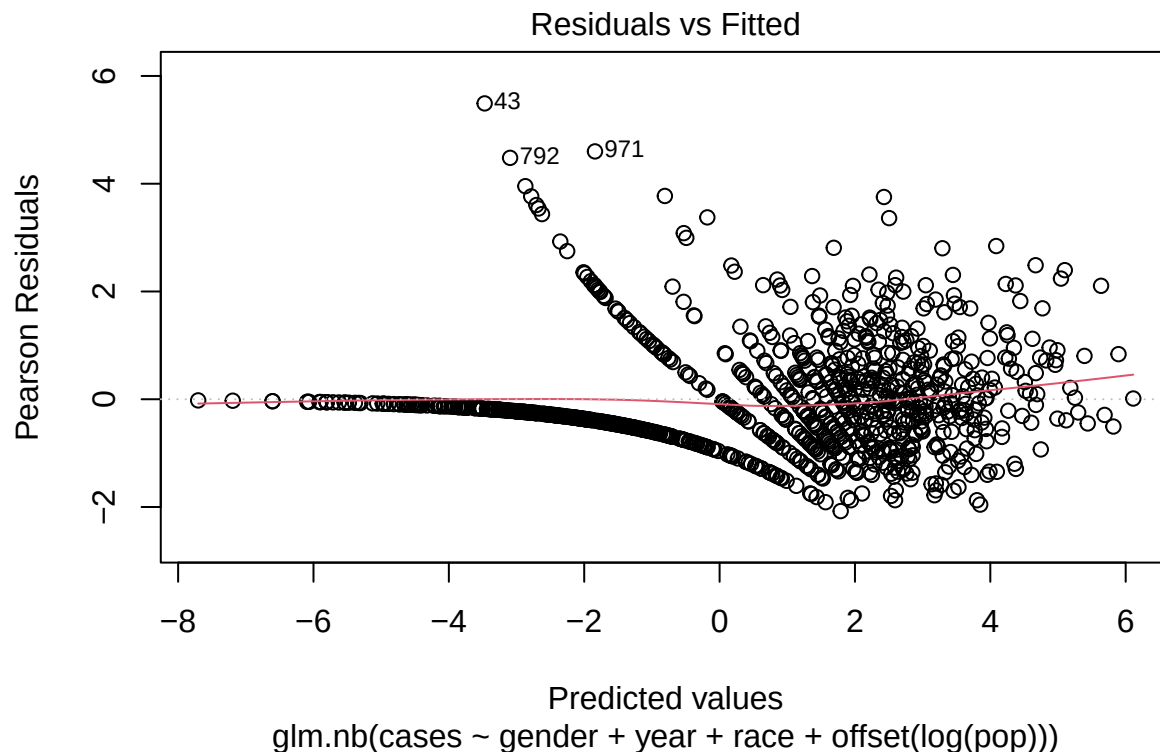


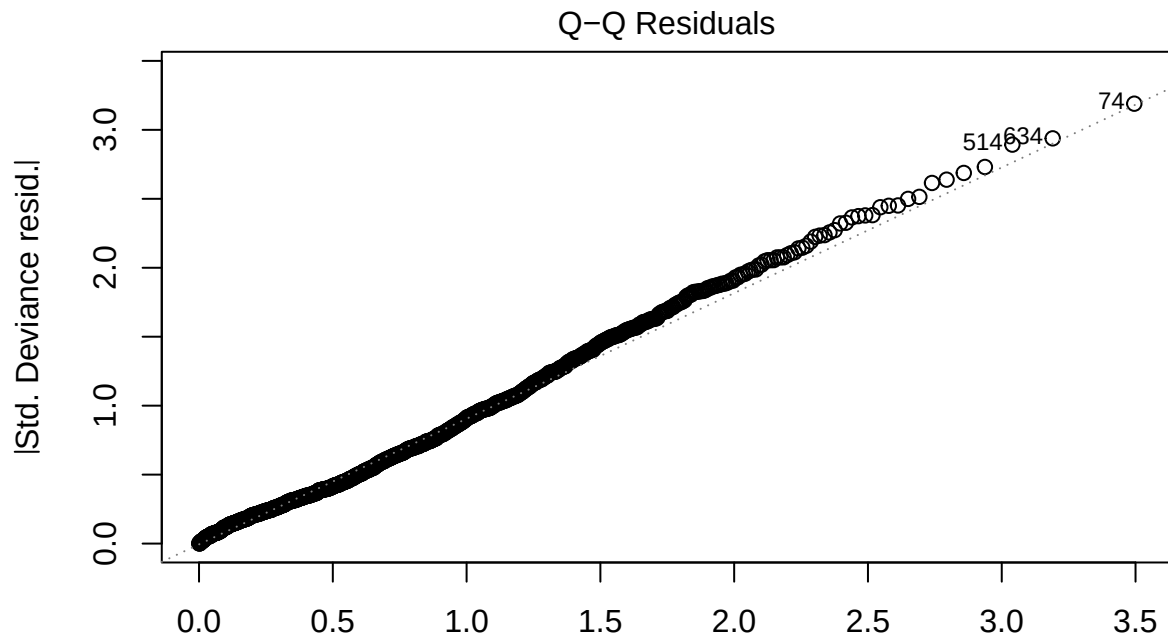
```
bnglm <- glm.nb(formula = cases ~ gender + year + race +
  offset(log(pop)),
  data=ohlung_long )
summary(bnglm)
```

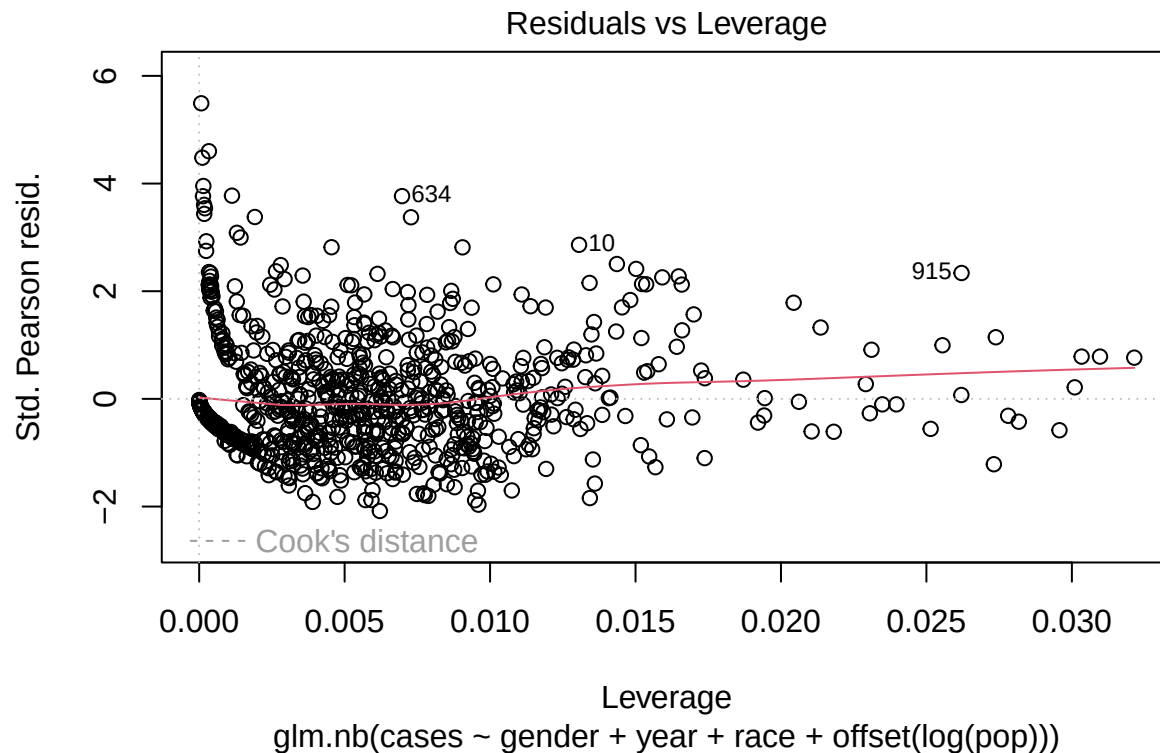
##

```
## Call:
## glm.nb(formula = cases ~ gender + year + race + offset(log(pop)),
##       data = ohlung_long, init.theta = 15.37679493, link = log)
##
## Coefficients:
##             Estimate Std. Error  z value Pr(>|z|)
## (Intercept) -8.80244    0.05379 -163.645  <2e-16 ***
## genderMale   1.11210    0.03265  34.057  <2e-16 ***
## year1978     0.43496    0.04123  10.550  <2e-16 ***
## year1988     0.78029    0.03981  19.602  <2e-16 ***
## raceWhite   -0.09919    0.04417  -2.246   0.0247 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(15.3768) family taken to be 1)
##
## Null deviance: 2459.18  on 1055  degrees of freedom
## Residual deviance:  916.48  on 1051  degrees of freedom
## AIC: 3970.8
##
## Number of Fisher Scoring iterations: 1
##
##              Theta:  15.38
##             Std. Err.:  1.87
##
## 2 x log-likelihood: -3958.77
```

```
plot(bnglm)
```



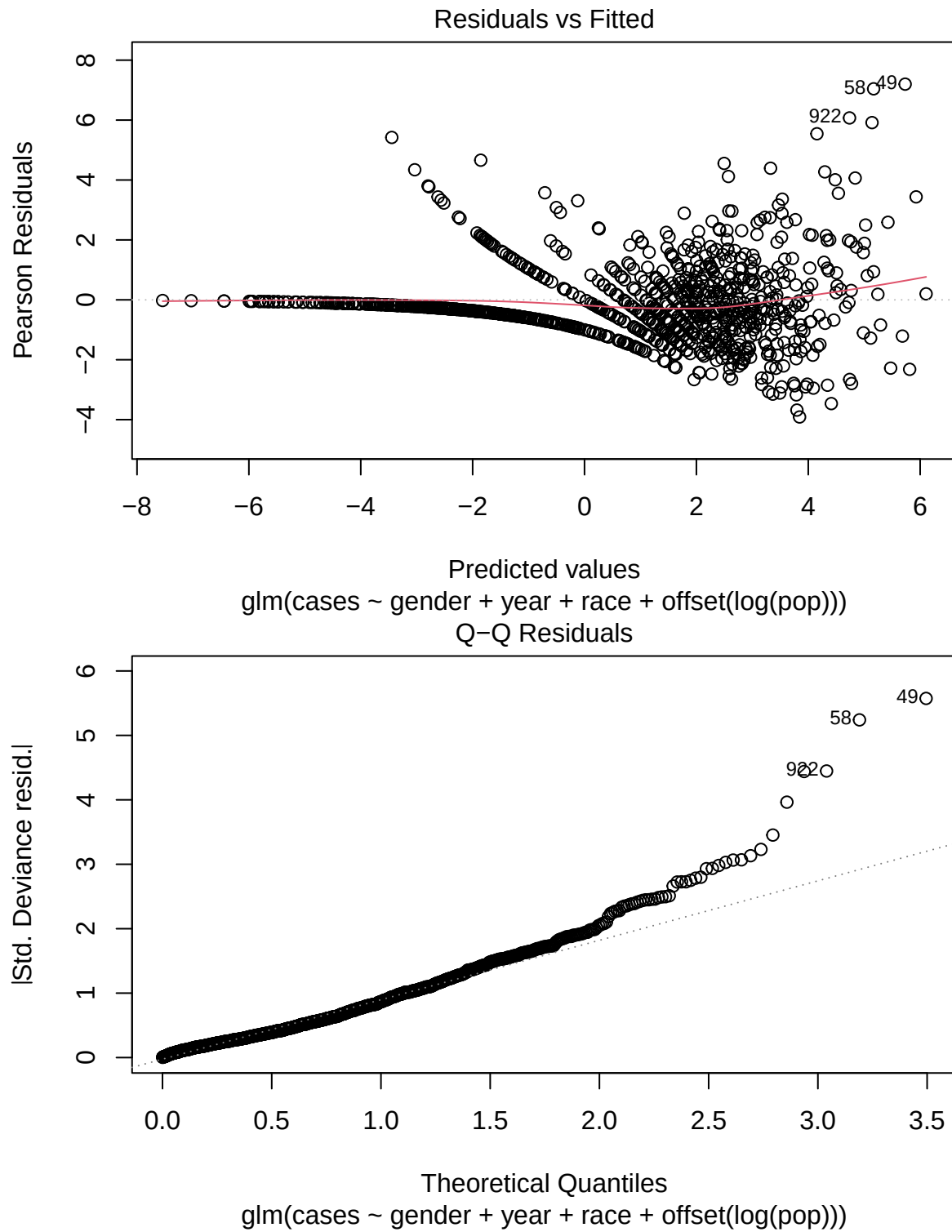


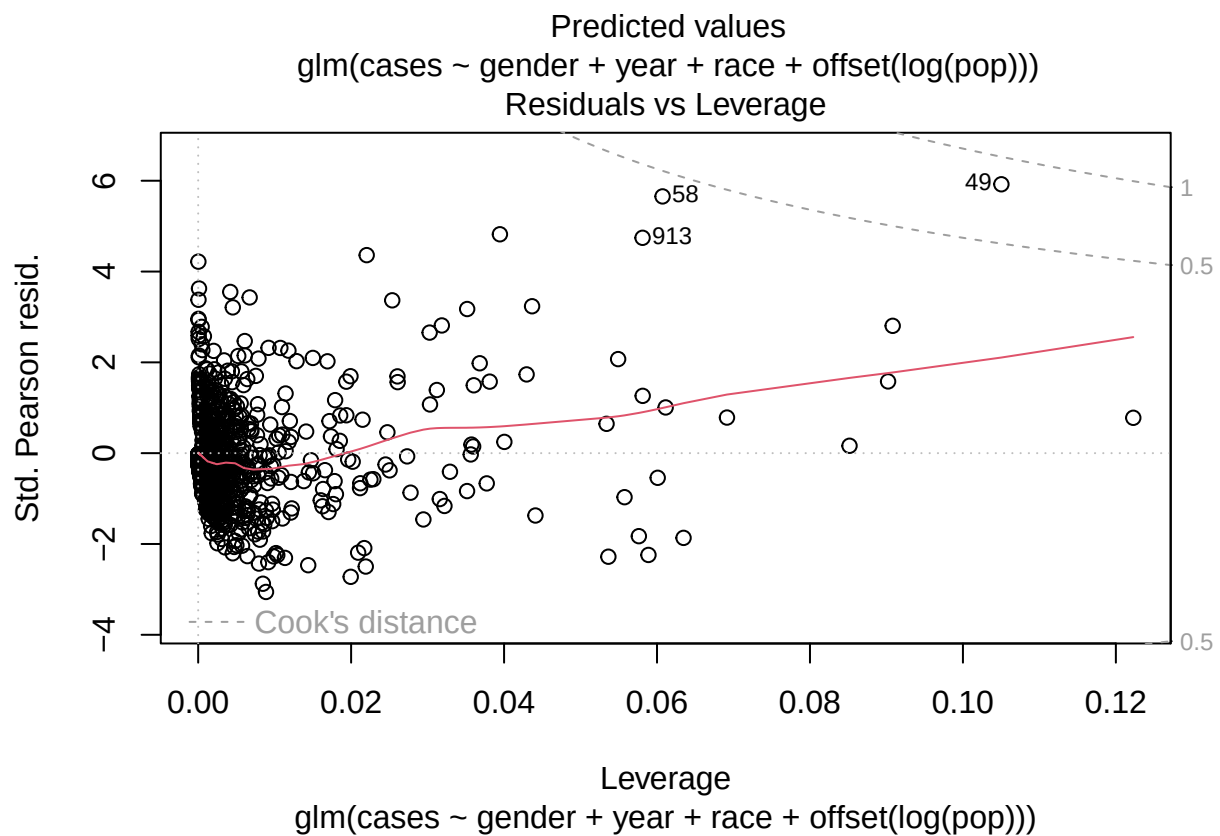
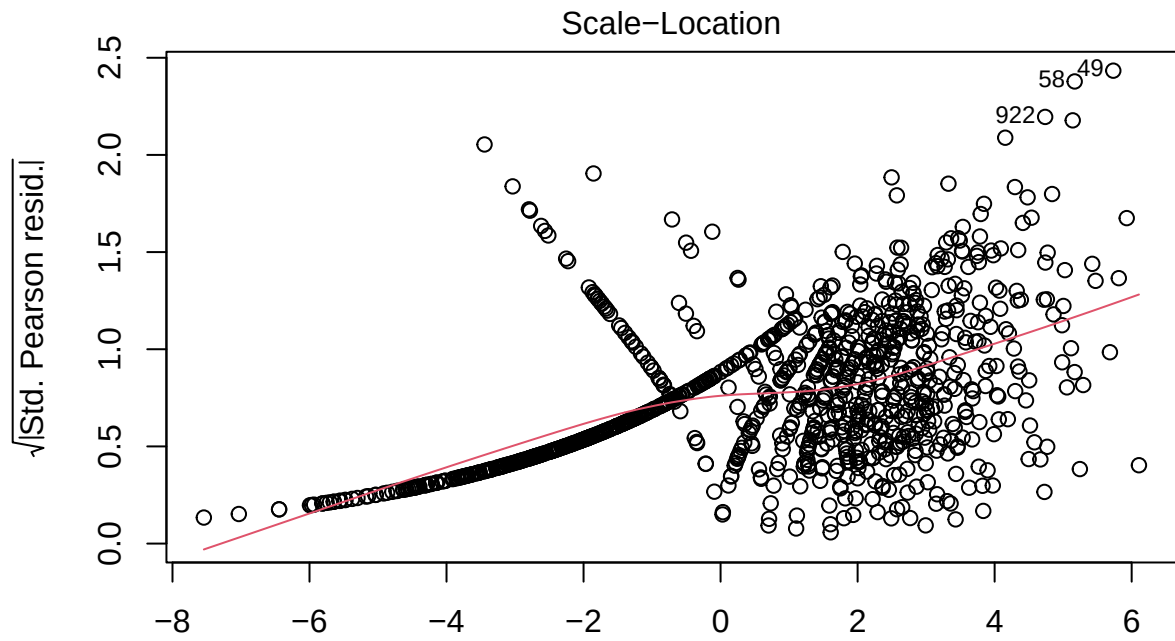


```
glm_qp <- glm(cases ~ gender + year + race + offset(log(pop)),
              family = quasipoisson(link = "log"),
              data = ohlung_long)
summary(glm_qp)
```

```
##
## Call:
## glm(formula = cases ~ gender + year + race + offset(log(pop)),
##      family = quasipoisson(link = "log"), data = ohlung_long)
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.64087    0.04127 -209.350 < 2e-16 ***
## genderMale   1.03832    0.02385  43.532 < 2e-16 ***
## year1978     0.37313    0.02920  12.780 < 2e-16 ***
## year1988     0.67867    0.02761  24.577 < 2e-16 ***
## raceWhite   -0.09129    0.03339  -2.734  0.00635 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 1.651373)
##
##      Null deviance: 6330.2  on 1055  degrees of freedom
## Residual deviance: 1731.9  on 1051  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 4
```

```
plot(glm_qp)
```



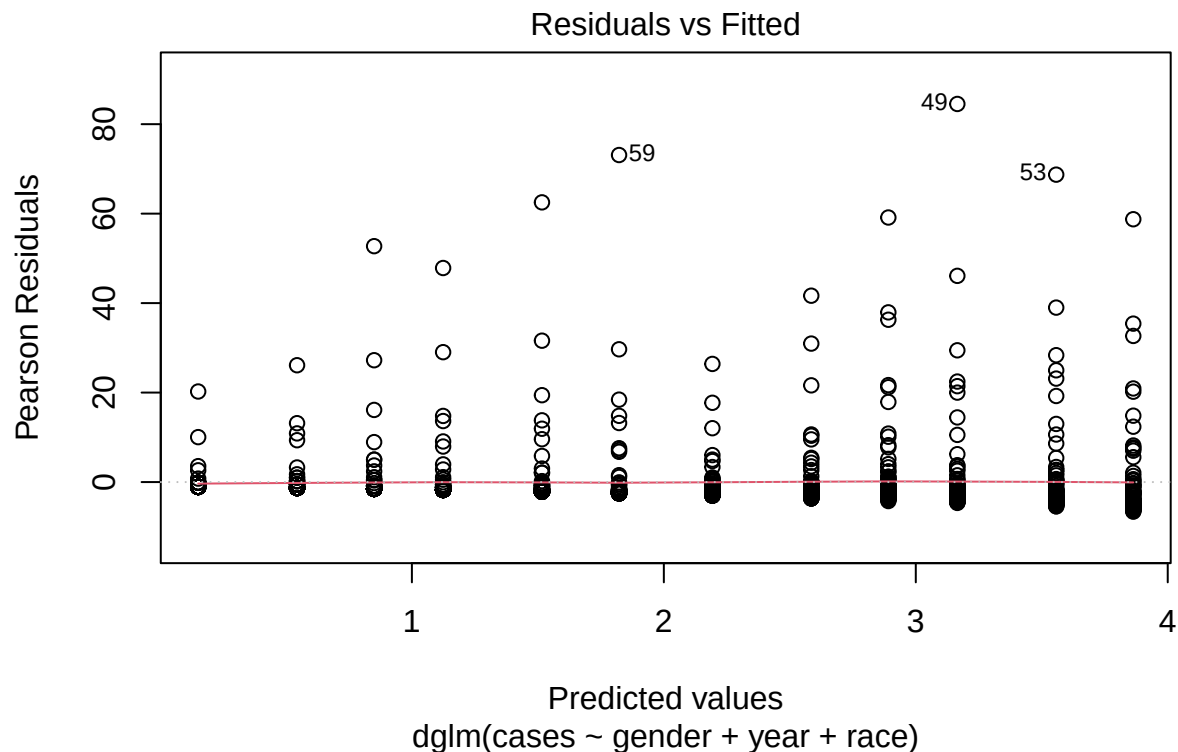


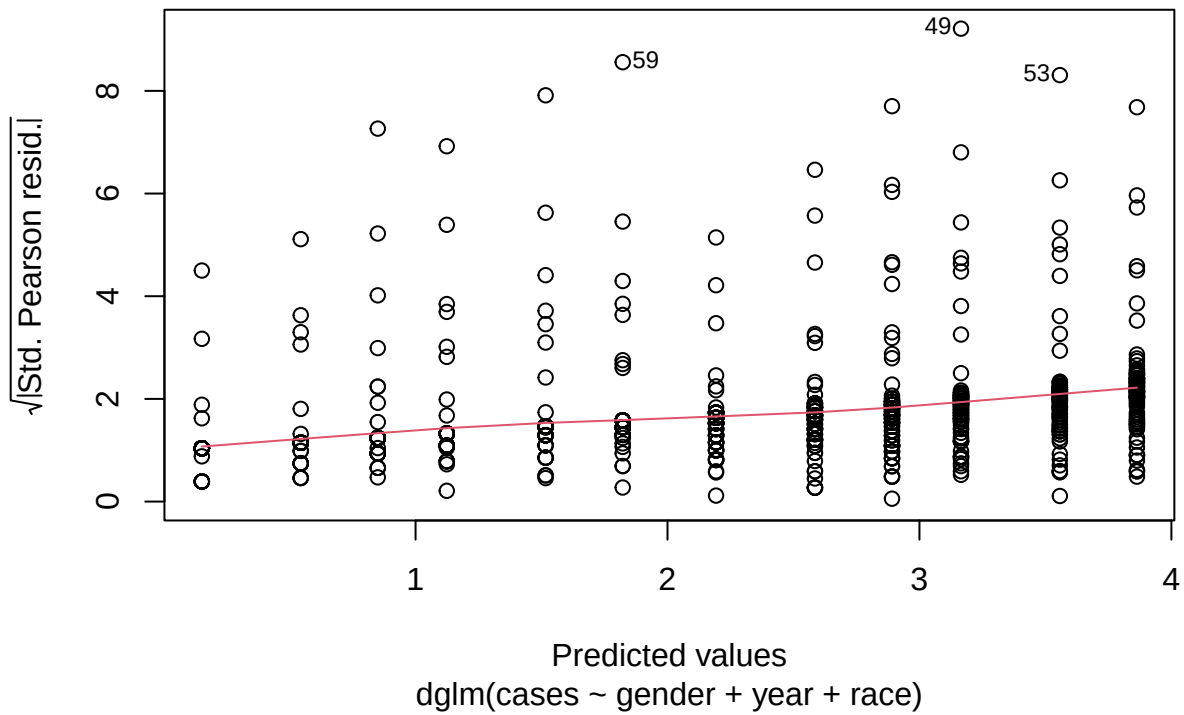
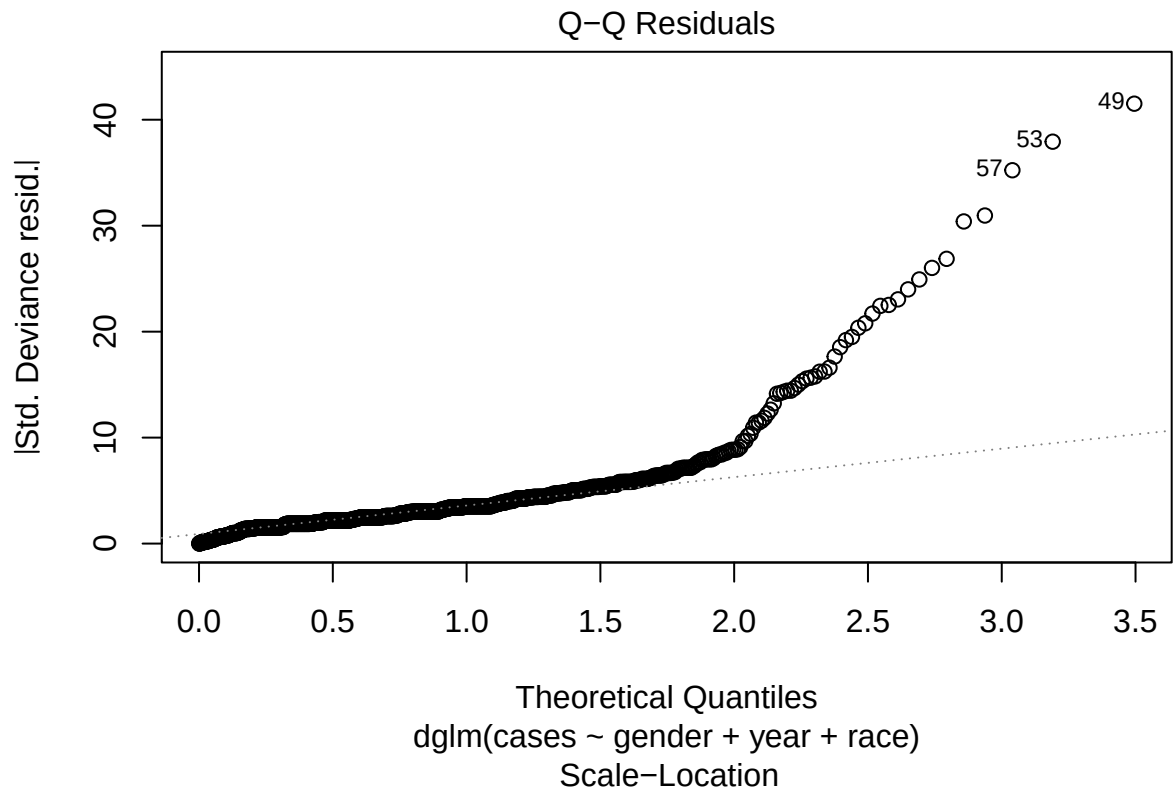
```
dglm_model <- dglm(cases ~ gender + year + race,
  offset = log(pop),
  family = poisson,
  data = ohlung_long)
```

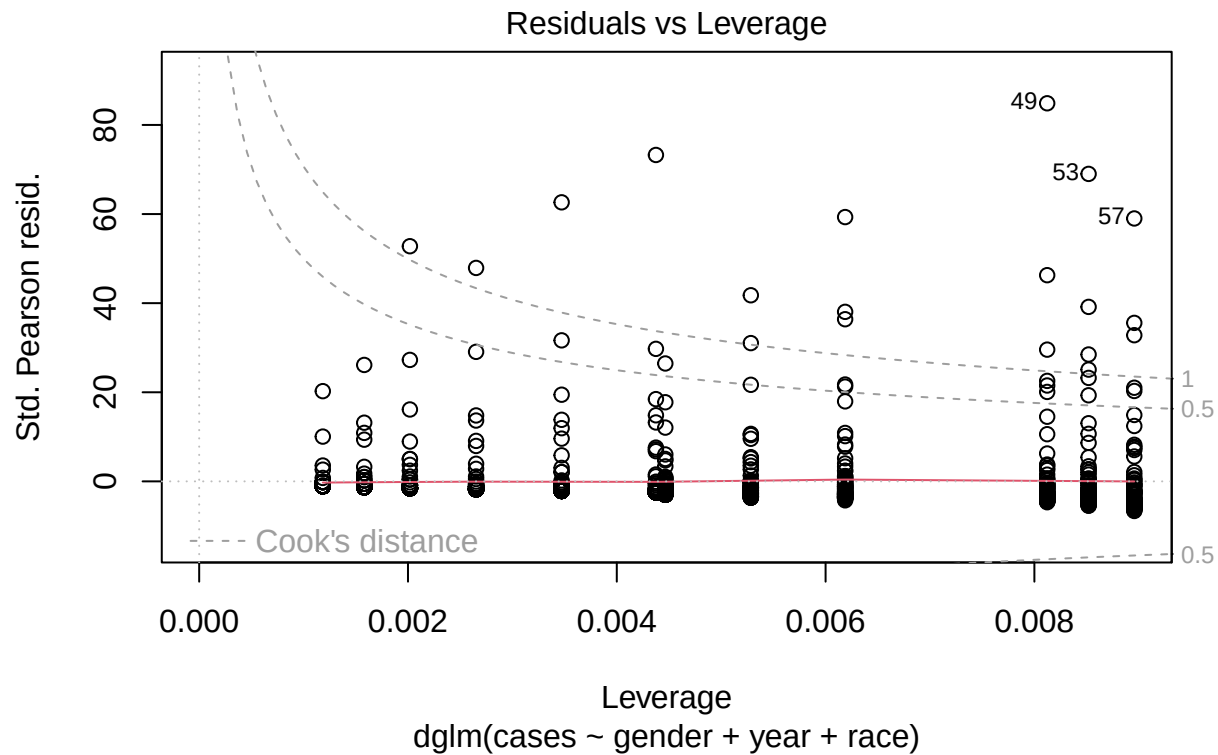
```
summary(dglm_model)
```

```
##
## Call: dglm(formula = cases ~ gender + year + race, family = poisson,
## data = ohlung_long, offset = log(pop))
##
## Mean Coefficients:
##           Estimate Std. Error      z value Pr(>|z|)
## (Intercept) 0.1517692 1.221981e+27 1.241993e-28      1
## genderMale 0.9721375 7.110272e+26 1.367230e-27      1
## year1978 0.3924417 8.703391e+26 4.509066e-28      1
## year1988 0.6983707 8.228045e+26 8.487687e-28      1
## raceWhite 2.0409723 9.946864e+26 2.051875e-27      1
## (Dispersion Parameters for poisson family estimated as below )
##
## Scaled Null Deviance: 0 on 1055 degrees of freedom
## Scaled Residual Deviance: 0 on 1051 degrees of freedom
##
## Dispersion Coefficients:
##           Estimate Std. Error  z value Pr(>|z|)
## (Intercept) 131.631 0.04351941 3024.649      0
## (Dispersion parameter for Gamma family taken to be 2 )
##
## Scaled Null Deviance: 1622.305 on 1055 degrees of freedom
## Scaled Residual Deviance: 136071.7 on 1055 degrees of freedom
##
## Minus Twice the Log-Likelihood: 141625.8
## Number of Alternating Iterations: 50
```

```
plot(dglm_model)
```







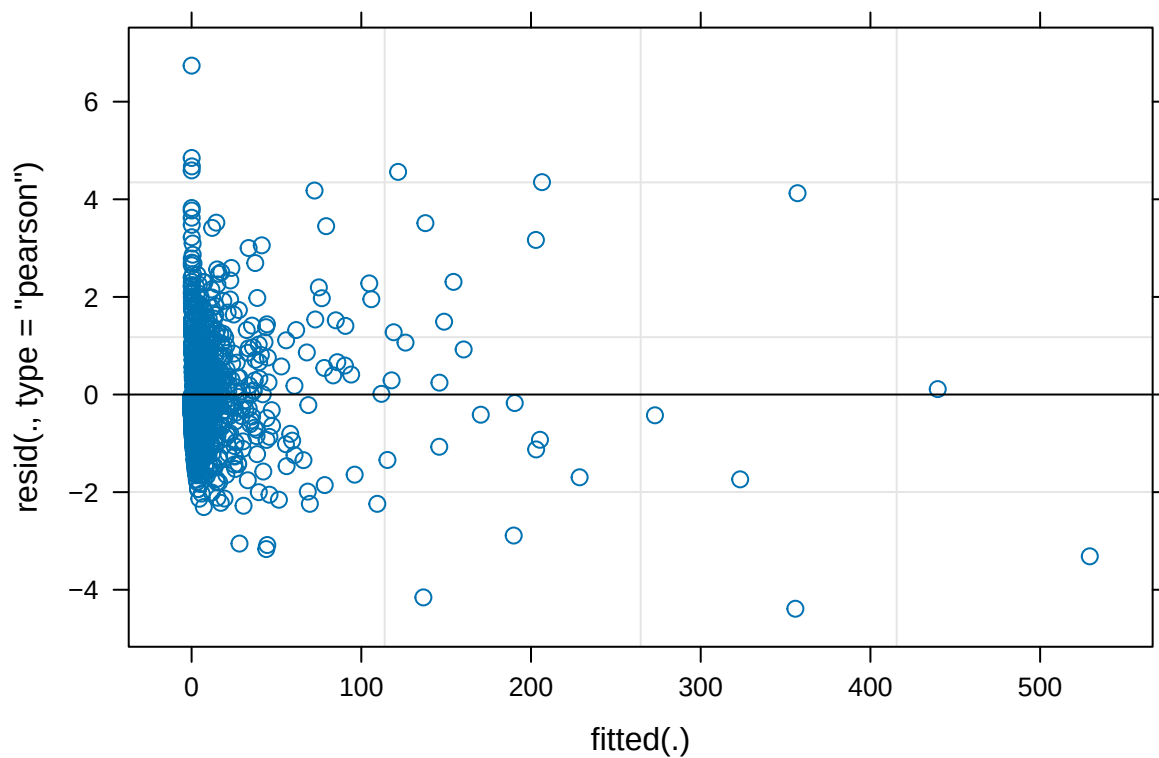
```
glmer <- glmer(cases ~ gender + race + year + (1 | NAME) +
  offset(log(pop)),
  family = poisson(link = "log"),
  data = ohlung_long)
```

```
summary(glmer)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: poisson ( log )
## Formula: cases ~ gender + race + year + (1 | NAME) + offset(log(pop))
## Data: ohlung_long
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  4046.2   4076.0   -2017.1   4034.2     1050
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.3887 -0.5452 -0.2137  0.3702  6.7391
##
## Random effects:
##  Groups Name      Variance Std.Dev.
##  NAME  (Intercept) 0.05086  0.2255
## Number of obs: 1056, groups: NAME, 88
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.833271   0.043562 -202.774  <2e-16 ***
## genderMale   1.041611   0.018546  56.164  <2e-16 ***
```

```
## raceWhite    -0.007412    0.026825   -0.276    0.782
## year1978      0.385094    0.022718   16.951   <2e-16 ***
## year1988      0.697595    0.021514   32.425   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) gndrMl racWht yr1978
## genderMale -0.309
## raceWhite  -0.604 -0.005
## year1978   -0.330  0.001  0.020
## year1988   -0.365  0.002  0.045  0.632
```

```
plot(glmer)
```



```
# Modelo con binomial negativa + efecto aleatorio por condado
nbea <- glmmTMB(cases ~ gender + race + year + (1 | NAME) +
                offset(log(pop)),
                family = nbinom2,           # + aquí está el cambio clave
                data = ohlung_long)

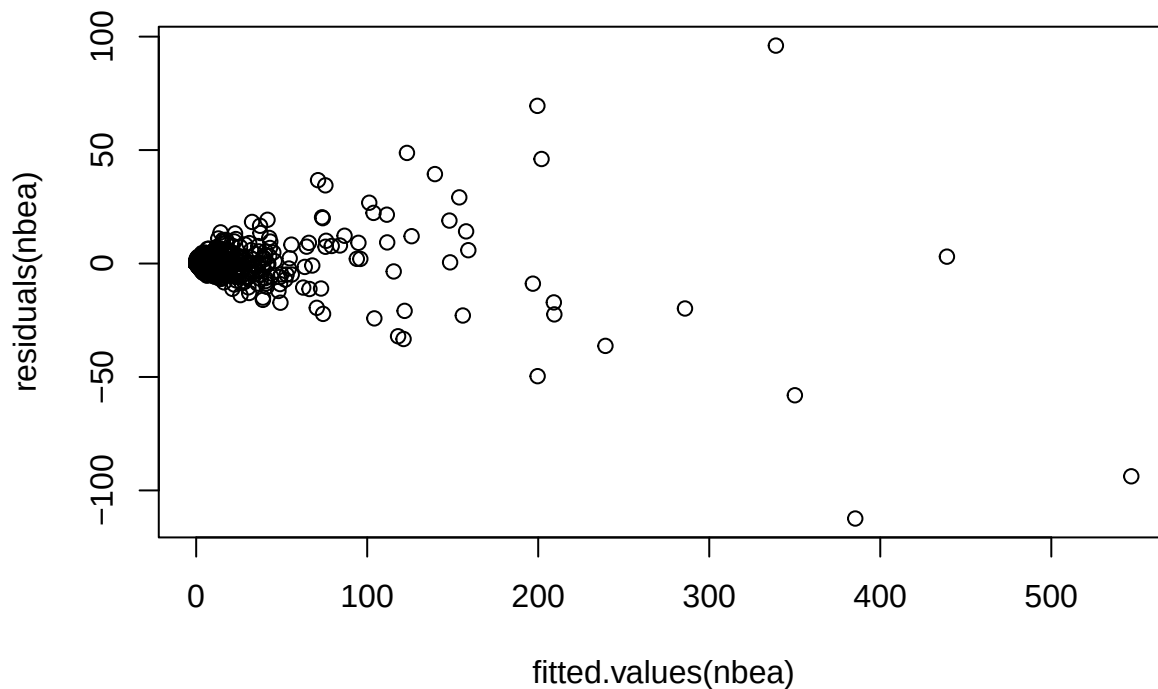
summary(nbea)
```

```
## Family: nbinom2 ( log )
## Formula:      cases ~ gender + race + year + (1 | NAME) + offset(log(pop))
## Data: ohlung_long
##
##           AIC          BIC      logLik -2*log(L)  df.resid
```



```
##      3884.9      3919.6     -1935.4      3870.9          1049
##
## Random effects:
##
## Conditional model:
##   Groups Name      Variance Std.Dev.
##   NAME      (Intercept) 0.04709  0.217
## Number of obs: 1056, groups:  NAME, 88
##
## Dispersion parameter for nbinom2 family (): 34.5
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.928147   0.058202 -153.40  <2e-16 ***
## genderMale   1.108472   0.028017  39.56  <2e-16 ***
## raceWhite   -0.008261   0.040879  -0.20    0.84
## year1978     0.435512   0.034816  12.51  <2e-16 ***
## year1988     0.782154   0.033656  23.24  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
plot(fitted.values(nbea),residuals(nbea))
```



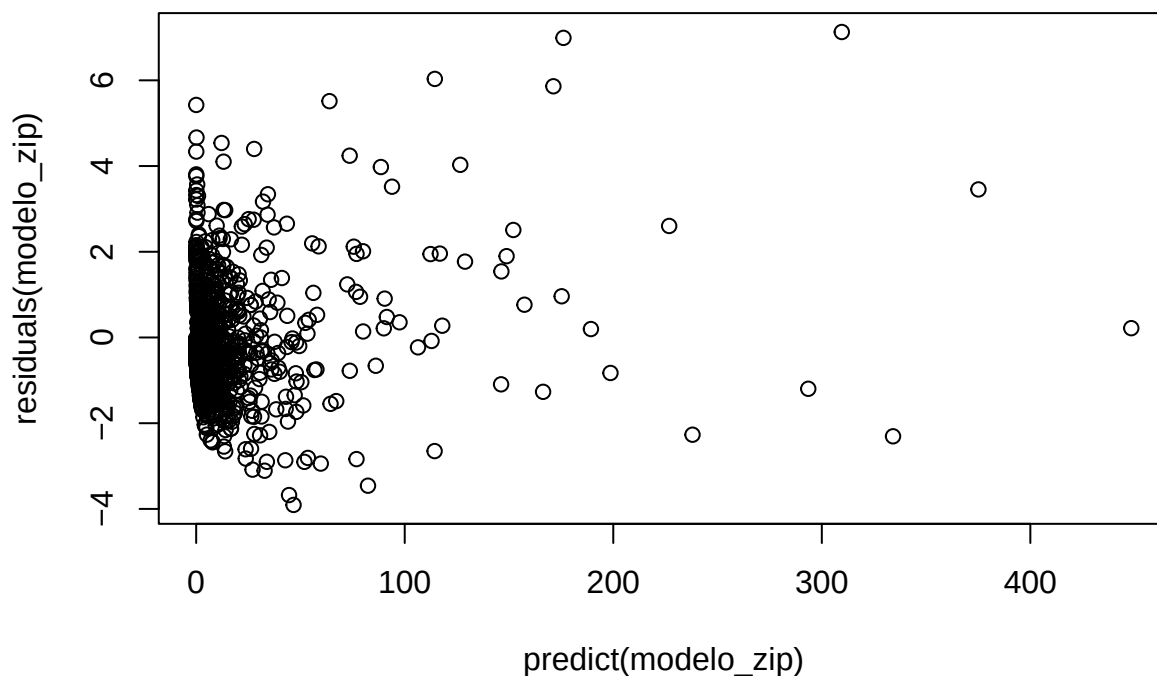
```
modelo_zip <- zeroinfl(
  cases ~ gender + race + year + offset(log(pop)),
  data = ohlung_long,
  dist = "poisson"
)
```

```
## Warning in value[[3L]](cond): sistema es computacionalmente singular: número de
## condición recíproco = 2.05739e-17FALSE
```

```
summary(modelo_zip)
```

```
##
## Call:
## zeroinfl(formula = cases ~ gender + race + year + offset(log(pop)), data = ohlung_long,
##   dist = "poisson")
##
## Pearson residuals:
##      Min       1Q   Median       3Q      Max
## -3.9065 -0.7381 -0.2869  0.2436  7.1273
##
## Count model coefficients (poisson with log link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -8.63504         NA      NA      NA
## genderMale    1.03490         NA      NA      NA
## raceWhite    -0.09034         NA      NA      NA
## year1978      0.36893         NA      NA      NA
## year1988      0.67449         NA      NA      NA
##
## Zero-inflation model coefficients (binomial with logit link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -32.98         NA      NA      NA
## genderMale   -13.24         NA      NA      NA
## raceWhite    18.76         NA      NA      NA
## year1978     -27.94         NA      NA      NA
## year1988     -10.11         NA      NA      NA
##
## Number of iterations in BFGS optimization: 45
## Log-likelihood: -2181 on 10 Df
```

```
plot(predict(modelo_zip),residuals(modelo_zip))
```



```

modelo_zinb <- zeroinfl(
  cases ~ gender + race + year + offset(log(pop)) | 1,
  data = ohlung_long,
  dist = "negbin"
)

summary(modelo_zinb)

```

```

##
## Call:
## zeroinfl(formula = cases ~ gender + race + year + offset(log(pop)) |
##      1, data = ohlung_long, dist = "negbin")
##
## Pearson residuals:
##      Min      1Q  Median      3Q      Max
## -2.0757 -0.5571 -0.2209  0.3470  5.4905
##
## Count model coefficients (negbin with log link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.80244    0.05432 -162.034 <2e-16 ***
## genderMale   1.11210    0.03269  34.024 <2e-16 ***
## raceWhite   -0.09919    0.04376  -2.267  0.0234 *
## year1978     0.43496    0.04091  10.631 <2e-16 ***
## year1988     0.78029    0.03969  19.659 <2e-16 ***
## Log(theta)   2.73286    0.12339  22.148 <2e-16 ***
##
## Zero-inflation model coefficients (binomial with logit link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -13.08      62.20  -0.21   0.833
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Theta = 15.3768
## Number of iterations in BFGS optimization: 31
## Log-likelihood: -1979 on 7 Df

```

```

plot(predict(modelo_zinb),residuals(modelo_zinb))

```

